

Study Report: The Definitive Guide to DAX - Power BI Analyst

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Family: Power BI Analyst
Level: Intermediate
Chapters: 7
Source: Reference: The Definitive Guide to DAX - Marco Russo & Alberto Ferrari

Official sources and free libraries

- <https://www.sqlbi.com/books/definitive-guide-to-dax/>

Summary

Study report for **The Definitive Guide to DAX** by Marco Russo & Alberto Ferrari (Intermediate level) mapped to the Power BI Analyst role. DAX measures, CALCULATE, time intelligence, and filter context.

Chapters

Track: Book-Intro

Chapter 1: About The Definitive Guide to DAX [Intermediate]

Chapter focus: About The Definitive Guide to DAX** *(Level: Intermediate)

This study report summarizes **The Definitive Guide to DAX** by Marco Russo & Alberto Ferrari for the Power BI Analyst role. The resource is rated Intermediate level. DAX measures, CALCULATE, time intelligence, and filter context. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# The Definitive Guide to DAX
# Author: Marco Russo & Alberto Ferrari
# Role: Power BI Analyst
# Level: Intermediate
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on The Definitive Guide to DAX.

When to use it

When learning Power BI Analyst skills at Intermediate level.

Where to use it

Power BI Analyst training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Intermediate]

Chapter focus: Why This Book Matters for Your Role** *(Level: Intermediate)

Power BI Analyst professionals use ideas from The Definitive Guide to DAX to solve real workplace problems. DAX measures, CALCULATE, time intelligence, and filter context. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: Power BI Analyst

Book focus: DAX measures, CALCULATE, time intelligence, and filter context.

Recommended level: Intermediate

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

Power BI Analyst bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a intermediate skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Intermediate]

Chapter focus: Key Topics Covered** *(Level: Intermediate)

The main topics in The Definitive Guide to DAX include practical concepts described as: DAX measures, CALCULATE, time intelligence, and filter context. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Power BI Analyst jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Advanced Power Query (M) Patterns [Intermediate]

Chapter focus: Advanced Power Query (M) Patterns** *(Level: Intermediate)

Parameterize file paths and server names for portability. Use List.Generate for pagination APIs; Table.NestedJoin for complex merges. Enable query folding - check View Native Query to push work to SQL.

Code Reference:

```
let
    Page = (n as number) =>
    Json.Document(Web.Contents("https://api.example.com/items?page=" & Text.From(n))),
    Pages = List.Generate(() => [i=1, data=Page(1)], each [data] <> null, each [i=[i]+1, data=Page([i]+1)])
in
```

```
Table.FromList(Pages, each [data][items])
```

What it shows: List.Generate loops API pages - common pattern for marketing APIs.

Topic guide

What it actually is

Advanced M handles APIs, parameters, and folding-aware transforms.

When to use it

Refreshing models from REST APIs or multi-file folders.

Where to use it

Marketing spend APIs, blob folders, and incremental file drops.

Real use example

Parameterized server name lets dev/test/prod switch without rewriting queries.

Chapter 5: Applied: DAX CALCULATE and Filter Context [Intermediate]

Chapter focus: DAX CALCULATE and Filter Context** *(Level: Intermediate)

CALCULATE modifies filter context - the heart of DAX. Sales YTD = TOTALYTD ([Total Sales], 'Date'[Date]). Understand row context vs filter context before iterator functions (SUMX).

Code Reference:

```
Sales CY = CALCULATE ( [Total Sales], 'Date'[Year] = 2025 )
Sales PY = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( 'Date'[Date] ) )
YoY % = DIVIDE ( [Sales CY] - [Sales PY], [Sales PY] )
```

What it shows: SAMEPERIODLASTYEAR shifts the date filter for year-over-year comparisons.

Topic guide

What it actually is

CALCULATE applies filters to measures dynamically.

When to use it

Time intelligence, conditional KPIs, and % of total metrics.

Where to use it

Finance close packs, retail comps, and subscription growth boards.

Real use example

YoY % measure works on any visual slice - region, product, or rep.

Chapter 6: Applied: Row-Level Security (RLS) [Intermediate]

Chapter focus: Row-Level Security (RLS)** *(Level: Intermediate)

RLS restricts data rows per audience. Define roles with DAX filters on tables: [Email] =

USERPRINCIPALNAME(). Test with View as Role before publish. Dynamic RLS via security bridge tables scales to thousands of users.

Code Reference:

```
[Region] = LOOKUPVALUE ( Users[Region], Users[Email], USERPRINCIPALNAME() )
```

What it shows: LOOKUPVALUE maps the logged-in user to their allowed region.

Topic guide

What it actually is

RLS enforces data access inside the semantic model.

When to use it

Multi-tenant SaaS analytics and regional sales reports.

Where to use it

Franchise dashboards, partner portals, and HR sensitive metrics.

Real use example

A store manager sees only their district - same report, different rows.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Intermediate]

Chapter focus: Study Plan and Practice *(Level: Intermediate)**

Finish The Definitive Guide to DAX with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Power BI Analyst. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes  
Week 2: Exercises  
Week 3: Mini project  
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a intermediate project screenshot.